12TH BIOLOGY · CHAPTER 13 · SOLVED EXERCISE

Thermoregulation and Osmoregulation

Complete textbook exercise: MCQs, Short Questions, Long Questions and Inquisitive Questions

SECTION 1: MULTIPLE CHOICE QUESTIONS

1. Which term covers the rest of the three?

(a) Osmoregulation (b) Excretion
(c) Thermoregulation (d) Homeostasis

Answer: (d) Homeostasis — Homeostasis is the maintenance of a stable internal environment. Thermoregulation, osmoregulation and excretion are its three important processes.

2. Marine fishes compensate for water loss by:

(a) Producing dilute urine (b) Drinking seawater
(c) Absorbing salt (d) Reducing thirst

Answer: (b) Drinking seawater — Marine fishes are hypotonic to seawater, so water leaves their bodies by osmosis. To make up this loss they drink seawater continuously.

3. How do freshwater fish primarily gain water?

(a) Drinking (b) Active transport
(c) Osmosis (d) Diffusion

Answer: (c) Osmosis — Freshwater fish are hypertonic to their environment, so water enters continuously by osmosis, mainly through the gills. They rarely drink water.

4. What is the primary heat source for endotherms?

(a) Sunlight (b) Internal metabolism
(c) Warm rocks (d) Convection

Answer: (b) Internal metabolism — Endotherms (birds and mammals) generate most of their body heat through their own internal metabolism.

5. What is the control center for human thermoregulation?

(a) Pituitary gland (b) Hypothalamus
(c) Adrenal gland (d) Medulla oblongata

Answer: (b) Hypothalamus — The hypothalamus of the brain works as the “thermostat”. It receives signals from thermoreceptors and sends messages to muscles and glands.

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6. Which waste product requires the most water for excretion?

(a) Ammonia (b) Urea
(c) Uric acid (d) Creatinine

Answer: (a) Ammonia — Ammonia is highly toxic. One gram of ammonia needs 500 mL of water, while urea needs 50 mL and uric acid only 1 mL.

7. Which nitrogenous waste is the least toxic and water-insoluble?

(a) Urea (b) Ammonia
(c) Uric acid (d) Nitrate

Answer: (c) Uric acid — Uric acid is even less toxic than urea. It is less soluble in water and forms precipitates, so it is excreted with very little water.

8. In which part of the body, the conversion of ammonia to urea takes place?

(a) Liver (b) Muscles
(c) Brain (d) Small intestine

Answer: (a) Liver — Ammonia is converted into urea in the liver. Urea is then excreted through the kidneys.

SECTION 2: SHORT QUESTIONS

1. Define osmoregulators and osmoconformers.

Osmoregulators are the animals that can regulate the concentrations of water and solutes in their body fluids. They keep their internal concentration constant even when the concentration of the environment changes. Most vertebrates of aquatic and terrestrial environments are osmoregulators.

Osmoconformers are the animals which allow the osmotic concentration of their body fluids to match their environment. They cannot maintain the concentration of water and solutes in their bodies, so they remain isotonic to their surroundings. Examples are all marine invertebrates, hagfishes, sharks and rays.

2. Define thermoregulation and explain its needs.

Thermoregulation is the homeostatic process by which organisms maintain their internal body temperature within a tolerable range, regardless of the temperature of the external environment.

Need of thermoregulation

- **Enzymes:** Enzymes work best at an optimal temperature. Below this temperature, enzyme–substrate collisions decrease and the metabolic rate drops. Well above it, enzymes denature and lose their function permanently.
- **Cell membranes:** The lipid bilayer of cell membranes is temperature-sensitive. At high temperatures, membranes can lose their selective permeability. At low temperatures, they become rigid and brittle.

3. What are the two types of animals on the basis of the ability to thermoregulate?

On the basis of the ability to thermoregulate, animals are of two types:

1. **Poikilotherms:** These animals are unable to maintain their body temperature within narrow limits. Their body temperature changes with the temperature of the environment, e.g., all invertebrates, fishes, amphibians and reptiles.
2. **Homeotherms:** These animals maintain a relatively constant internal body temperature, regardless of environmental changes, e.g., birds and mammals. Humans keep their temperature at about 37 °C.

4. Differentiate between:

• Osmoregulators and osmoconformers

Osmoregulators	Osmoconformers
They can regulate the concentrations of water and solutes in their body fluids.	They cannot maintain the concentrations of water and solutes in their body fluids.
Their internal concentration stays constant even when the environment changes.	Their internal concentration changes with the surrounding medium, so they remain isotonic to it.
They use mechanisms such as active transport of salts, accumulation of salts and water conservation.	They do not use such mechanisms. They live in environments with stable solute concentration, such as marine water.
Examples: most vertebrates of aquatic and terrestrial environments.	Examples: all marine invertebrates, some freshwater invertebrates, hagfishes, sharks and rays.

• Ectotherms and endotherms

Ectotherms	Endotherms
They gain most of their body heat from external sources.	They generate most of their body heat through internal metabolism.
Their metabolic rate is very low, so they cannot produce enough heat.	Their metabolism produces enough heat to keep the body warm.
Their body temperature generally fluctuates with the surroundings.	Their body temperature is relatively constant and is often higher than the environment.
They use behavioural thermoregulation, e.g., basking in the sun or seeking shade.	They shiver or burn brown adipose tissue for heat, and have insulation such as hair, feathers or fat.
Examples: most invertebrates, fishes, amphibians and reptiles.	Examples: birds and mammals.

• Vasodilatation and vasoconstriction

Vasodilatation	Vasoconstriction
It is the expansion (widening) of the blood capillaries lying just beneath the epidermis of the skin.	It is the narrowing of the blood capillaries of the skin.
It occurs when the body temperature rises above 37 °C (hyperthermia).	It occurs when the body temperature falls below 37 °C (hypothermia).
More blood flows to the skin.	Less blood flows to the skin.
More heat is removed from the blood, so the body cools down.	Less heat is lost from the blood, so heat is saved in the body.

• Homeotherms and poikilotherms

Homeotherms	Poikilotherms
They maintain a relatively constant internal body temperature, regardless of environmental changes.	They are unable to maintain their body temperature within narrow limits.
Their body temperature does not follow the temperature of the environment.	Their body temperature varies considerably and usually matches the environment.
They use a “thermostat” (hypothalamus in the brain) to start cooling or heating responses.	They do not have such an effective internal control of temperature.
They need a large amount of food to keep the temperature constant.	They do not need massive amounts of food, so they can survive in nutrient-poor environments.
Examples: mammals and birds.	Examples: all invertebrates, fishes, amphibians and reptiles.

• Ammoniotelic and ureotelic

Ammoniotelic (ammonotelic) animals	Ureotelic animals
They excrete ammonia (NH_3) as their major nitrogenous waste.	They excrete urea, $CO(NH_2)_2$, as their major nitrogenous waste.
Ammonia is highly toxic and is excreted rapidly.	Urea is less toxic than ammonia.
1 g of ammonia needs 500 mL of water to dilute it to the non-toxic level.	1 g of urea needs 50 mL of water to dilute it to the non-toxic level.
They live where plenty of water is available, e.g., most fishes, protozoans, sponges and coelenterates.	They are the animals that must conserve water, e.g., terrestrial animals such as humans (mammals).

• Ureotelic and uricotelic

Ureotelic animals	Uricotelic animals
They excrete urea, $CO(NH_2)_2$, as their major nitrogenous waste.	They excrete uric acid ($C_5H_4N_4O_3$) as their nitrogenous waste.
Urea is less toxic than ammonia and dissolves in water.	Uric acid is even less toxic than urea. It is less soluble in water and forms precipitates.
1 g of urea needs 50 mL of water for excretion.	1 g of uric acid needs just 1 mL of water for excretion.
Ammonia is converted into urea in the liver.	Ammonia is converted into uric acid by using a lot of energy.
Example: terrestrial animals such as humans (mammals).	Examples: insects, desert animals, birds and most reptiles.

SECTION 3: LONG QUESTIONS

1. Describe three elements which operate homeostatic mechanisms.

Homeostasis is the process by which living organisms maintain a stable internal environment, despite changes in their external environment. It is maintained by the joint working of three essential elements: receptors, control centre and effectors.

1. Receptors

Receptors or sensors are specialized structures (organs, tissues, cells or subcellular components) that detect changes in the internal or external environment. They also transmit the information of these changes to the control centre. For example, receptor cells in the skin detect changes in temperature, and receptor cells in the blood vessels detect changes in blood pressure.

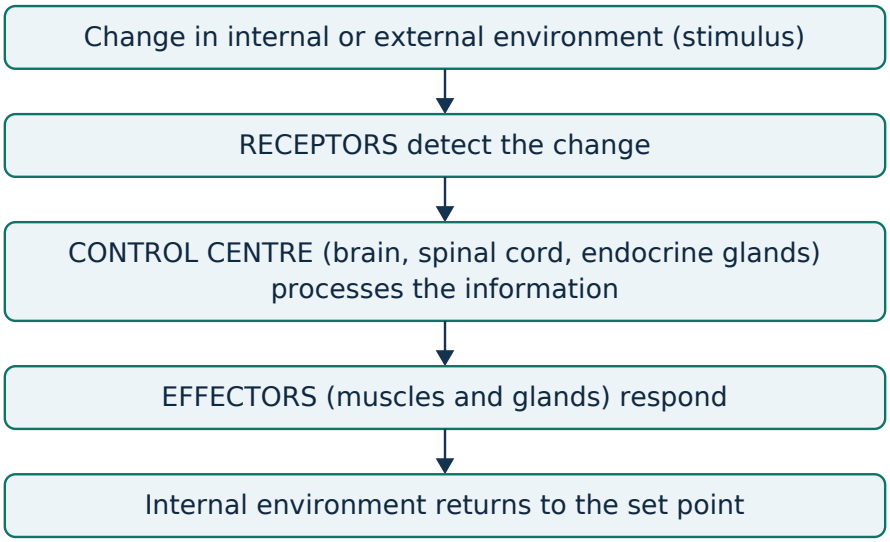
2. Control centre

The control centre receives information from the receptors, processes it and sends an appropriate message to the effectors. These messages keep the internal environment within a narrow range of **set points**, which are optimal for the working of the body. The brain, spinal cord and endocrine glands act as control centres.

3. Effectors

Effectors are the parts of the body which respond when they receive a message from the control centre. Muscles and glands act as effectors. Muscles respond by contracting, while glands release their secretions. In this way, the responses of the effectors bring the internal environment back to its set point.

Example: When the body temperature drops below the set point, the effectors in the skin and muscles cause shivering and vasoconstriction. This increases heat production and raises the body temperature back to the set point.



Elements of homeostasis

2. Relate the homeostatic mechanisms with the negative and positive feedback systems.

Feedback mechanisms are the foundations of homeostasis. They are the regulatory processes in which the output or product of a system is used to control the behaviour of the system itself. There are two types of feedback mechanisms.

Negative feedback

In negative feedback, the output or product of a system **reduces** the activity of the system. It is the most common form of regulation. Examples are the regulation of body temperature, blood glucose level and hormone secretion.

Example: If the body temperature rises above the set point of 37 °C, the thermoreceptors in the hypothalamus detect this change and send signals to the control centre (thermoregulator in the hypothalamus). The control centre sends messages to the effectors, e.g., sweat glands. They produce sweat, which lowers the temperature. This output (fall of temperature to the set point) affects the thermoreceptors negatively, so they stop sending messages to the control centre.

Positive feedback

In positive feedback, the output or product **increases** the activity of the system, so the original stimulus is enhanced. It is less common than negative feedback. It occurs in processes that need rapid and significant changes, e.g., blood clotting, labour and childbirth.

Example: During childbirth, contractions of the uterus send nerve impulses to the control centre (a part of the hypothalamus). It signals the pituitary gland to release oxytocin hormone. Oxytocin stimulates further contractions of the uterus, which send more impulses to the hypothalamus, and more oxytocin is released. This loop continues until the baby is born.

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Unregulated positive feedback

Sometimes an unregulated positive feedback causes health problems. For example, in severe inflammation or infection, immune cells release proteins called cytokines, which further stimulate the immune response. If this is not properly regulated, it leads to excessive inflammation and tissue damage.

3. Explain the different methods of osmoregulation found in freshwater, marine water and terrestrial habitats.

Osmoregulation is the maintenance of the balance of water and solutes in the body fluids. Animals of different habitats face different problems, so they use different methods.

Osmoregulation in freshwater

Problem: Freshwater animals are generally hypertonic to their environment, i.e., their body fluids have a higher solute concentration than the surrounding water. Water continuously enters their bodies by osmosis, especially across permeable surfaces like gills. Essential salts and ions (like Na^+ and Cl^-) diffuse out into the water.

Methods:

- They produce large volumes of dilute urine to remove the excess water.
- They actively transport salts back into their bodies to make up the loss of salts.
- They rarely drink water.

Osmoregulation in marine water

Problem: Most marine animals are hypotonic to their environment, i.e., seawater has a much higher salt concentration than their tissues. Water constantly leaves their bodies by osmosis through the gills and skin. Salts continuously diffuse into their bodies, and more salts enter by drinking seawater.

Methods:

- They drink seawater continuously to make up the water loss.
- They conserve water by producing very small amounts of highly concentrated urine.
- Specialized chloride cells in the gills use energy (ATP) to actively pump salts out of the blood into the environment.
- They also remove excess salts through their kidneys.
- Some marine fishes have salt-excreting rectal glands in the rectum. These glands pass salts from the blood into the digestive tract, and the salts leave with the faeces.

Osmoregulation in terrestrial habitats

Problem: Terrestrial animals lose water by evaporation from respiratory surfaces, through the skin and in the excretion of wastes. Their water intake (drinking and food) plus metabolic water must be equal to the water lost.

Methods: They have anatomical, physiological and behavioural adaptations.

- **Impermeable outer layers:** Insects have a waxy exoskeleton made of chitin. Reptiles have waterproof scales made of keratin. Mammals and birds have dead, keratinized skin cells and oil-secreting glands in the skin.

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- **Kidneys:** Their kidneys play a vital role in retaining water inside the body.
- **Nature of waste:** They convert ammonia into urea or uric acid, which need less water for excretion.
- **Behaviour:** Many desert animals stay underground during the heat of the day and come out only at night, when the air is cooler and more humid.
- **Metabolic water:** Desert animals eat seeds rich in carbohydrates. Some, like the kangaroo rat, survive entirely on metabolic water produced in cellular respiration.

4. Explain the nature of nitrogenous wastes in relation to habitat.

In many animals, osmoregulation is coupled with the removal of metabolic wastes. The main metabolic wastes are the nitrogenous wastes, i.e., ammonia, urea and uric acid. They are excreted dissolved in water. So the form of nitrogenous waste depends on how much water is available in the habitat of the animal.

1. Ammonia (NH_3) — ammonotelic animals

Ammonia is formed in the liver when the amino group (–NH_2) is removed from amino acids and nucleic acids and combines with H^+. It is highly toxic, so it is excreted rapidly. One gram of ammonia needs 500 mL of water to dilute it to the non-toxic level. Such plenty of water can only be afforded by freshwater animals, e.g., most fishes, protozoans, sponges and coelenterates. Marine animals excrete ammonia directly into the water. Animals which excrete ammonia as their major nitrogenous waste are called **ammonotelic**.

2. Urea, $CO(NH_2)_2$ — ureotelic animals

Terrestrial animals must conserve water, so they cannot afford to excrete ammonia with plenty of water. They convert ammonia into urea in the liver and excrete it through the kidneys. Urea is less toxic than ammonia. One gram of urea needs 50 mL of water. Animals which excrete urea as their major nitrogenous waste are called **ureotelic**.

3. Uric acid ($C_5H_4N_4O_3$) — uricotelic animals

Insects, desert animals, birds and most reptiles face an extreme challenge of water conservation because of their arid environment. They cannot afford to excrete even urea. They convert ammonia into uric acid by using a lot of energy. Uric acid is even less toxic than urea, is less soluble in water and forms precipitates. One gram of uric acid needs just 1 mL of water. Animals which excrete uric acid are called **uricotelic**.

Waste	Toxicity	Water for 1 g	Habitat / examples
Ammonia	Highly toxic	500 mL	Plenty of water: most fishes, protozoans, sponges, coelenterates
Urea	Less toxic	50 mL	Terrestrial animals that conserve water
Uric acid	Least toxic	1 mL	Arid habitats: insects, desert animals, birds, most reptiles

5. Classify animals on the basis of the source of body's heat.

On the basis of the source of body heat, animals are classified into ectotherms and endotherms.

1. Ectotherms

Ectotherms are the animals that gain most of their body heat from external sources. Their internal temperature generally fluctuates with the temperature of their surroundings. Most invertebrates are ectotherms. Among vertebrates, fishes, amphibians and reptiles are ectotherms.

Their metabolic rate is very low, so they cannot produce enough heat to raise and maintain their body temperature. They rely on heat from the environment and use **behavioural thermoregulation**. When they are too cold, they bask in the sun (radiation) or sit on a warm rock (conduction). When they are too hot, they seek shade or burrow into the cool earth. In very cold climates, they become dormant during the coldest months.

2. Endotherms

Endotherms generate the majority of their body heat through internal metabolism. They maintain a relatively constant body temperature, which is often higher than that of the environment. Birds and mammals are endotherms.

When it gets cold, they may shiver (muscle contractions) or burn brown adipose tissue to produce heat. Most endotherms also have insulation, such as hair, feathers or fat, to retain heat.

6. Classify the animals on the bases of the ability to thermoregulate.

On the basis of the ability to thermoregulate, animals are classified into poikilotherms and homeotherms.

1. Poikilotherms

Poikilotherms are the animals that are unable to maintain their body temperature within narrow limits. Their body temperature varies considerably and usually matches the temperature of the environment. All invertebrates are poikilotherms. Among vertebrates, fishes, amphibians and reptiles are poikilotherms.

Poikilotherms do not need to consume massive amounts of food to "fuel" a constant temperature. So they can survive in nutrient-poor environments where a homeotherm would starve.

2. Homeotherms

Homeotherms maintain a relatively constant internal body temperature, regardless of environmental fluctuations. They use a "thermostat" (the hypothalamus in the brain) to trigger cooling or heating responses. Mammals and birds are homeotherms. For instance, humans maintain their internal temperature at roughly 37 °C whether they are in a very cold or a very hot environment.

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7. Describe the regulatory strategies in man for thermoregulation.

Humans are endothermic homeotherms. They regulate their body temperature through a negative feedback mechanism. Thermoreceptors in the skin detect an increase or decrease in body temperature and send signals to the control centre, i.e., the hypothalamus, which also contains thermoreceptors. The hypothalamus sends signals to muscles and glands, which produce the corrective responses.

Responses to overheating (hyperthermia)

When the body temperature rises above 37 °C:

- **Vasodilation:** The blood capillaries lying just beneath the epidermis of the skin expand. More blood flows in them, so more heat is removed from the blood.
- **Sweating:** Sweat glands produce sweat. Evaporation of sweat carries heat away from the body and produces a cooling effect.
- **Lower metabolic rate:** The thyroid gland may decrease the secretion of thyroxine. This lowers the basal metabolic rate, so internal heat production is reduced.

Responses to cooling (hypothermia)

When the body temperature falls below 37 °C:

- **Vasoconstriction:** The skin capillaries become narrow. Blood flow to the skin is reduced, so less heat is lost from the blood.
- **Shivering:** Spasmodic contraction of skeletal muscles produces heat, which helps to raise the body temperature.
- **Goosebumps:** Tiny muscles at the base of hair follicles contract and the hair stand erect. Air is trapped in the erected hair and acts as an insulator.
- **Higher metabolic rate:** An increased metabolic rate produces more heat.

Behavioural responses

The thermoreceptors also stimulate the higher centres of the brain, so we “feel” warmth or coldness. This allows us to take further actions, like taking a bath or turning on the fans or heaters.

INQUISITIVE QUESTIONS

1. Why positive feedback mechanisms in humans are sometimes associated with severe health problems?

In positive feedback, the output of a system increases the activity of the same system. So the change does not return to the set point. It keeps growing stronger and stronger until something stops it. In childbirth it stops naturally when the baby is born. But if a positive feedback loop is not regulated, the response becomes excessive and harms the body.

For example, in severe inflammation or infection, immune cells release proteins called cytokines. These cytokines further stimulate the immune response, so even more cytokines are released. If this loop is not properly controlled, it leads to excessive

inflammation and tissue damage. That is why positive feedback is less common in the body than negative feedback.

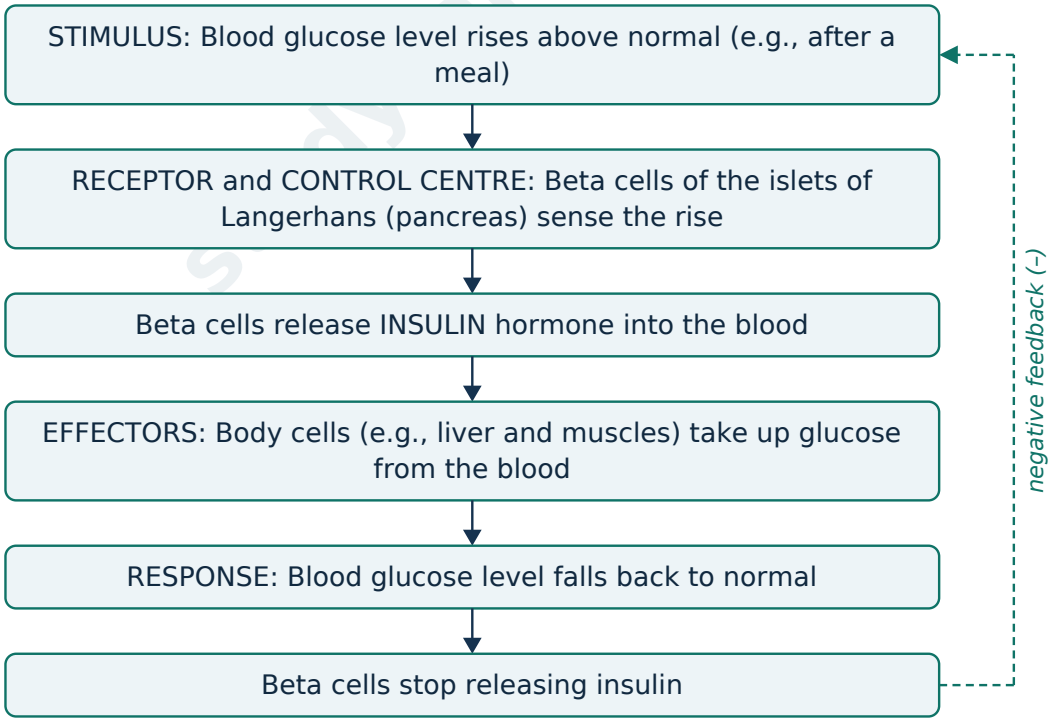
2. List some of the behavioural responses of the animals to maintain homeostasis.

Animals show the following behavioural responses to maintain homeostasis:

- Ectotherms bask in the sun or sit on a warm rock when they are too cold.
- When they are too hot, they seek shade or burrow into the cool earth.
- In very cold climates, ectotherms become dormant during the coldest months.
- Many desert animals stay underground during the heat of the day and come out only at night to avoid loss of water by evaporation.
- Marine fishes drink seawater continuously to make up their water loss, while freshwater animals rarely drink water.
- Humans take a bath or turn on fans when they feel hot, and turn on heaters when they feel cold.

3. Draw a flow chart to show negative feedback of homeostatic mechanisms by taking an example of hormone.

Example: control of blood glucose level by insulin hormone. When the blood glucose level rises above normal, insulin is released. Insulin lowers the blood glucose level to normal. This result (normal glucose level) stops the further release of insulin. So the product of the process stops the process itself, which is negative feedback.



Negative feedback in the control of blood glucose level by insulin